Bivariate normal distribution parameters and "geometry"

$(X_1,X_2) \sim N(\mu,\Sigma)$

$\mu = [0, 0]$ $\quad [\sigma_1, \sigma_2] = [1, 1]$ $\quad \rho = 0$

$\Sigma = [1\ \ 0; 0\ \ 1]$

total variation $= \Sigma_i \lambda_i = \sigma_1^2 + \sigma_2^2 = 2$

generalized variance $= |\Sigma| = \Pi_i \lambda_i = \sigma_1^2 * \sigma_2^2 * (1 - \rho^2) = 1$

---

constant density contour ellipses $(x-\mu)^T * \Sigma^{-1} * (x-\mu) = c^2$

eigenvalues and vectors for Σ $\quad \lambda_1 = 1 \quad e_1 = [1\ \ 0]$

$\lambda_2 = 1 \quad e_2 = [0\ \ 1]$

Mahalanobis (statistical) distance $(x-\mu)^T * \Sigma^{-1} * (x-\mu) \sim \chi^2_2$

---

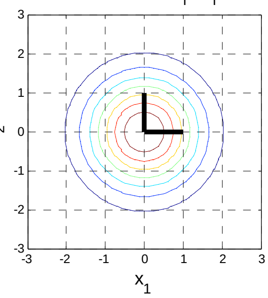
therefore $c^2 = \chi^2_{2,\alpha}$ gives $(1-\alpha)$*100% prediction ellipses for observations, x_i

ellipse half-length axes $\sqrt{\lambda_i} * \sqrt{\chi^2_{2,\alpha}} * e_i = [1 * e_1,\ 1 * e_2] * \sqrt{\chi^2_{2,\alpha}}$

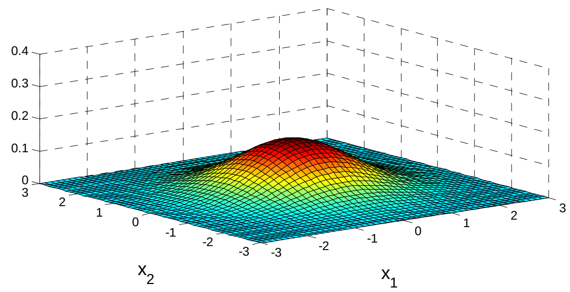
ellipse areas $A_\alpha = \pi * \chi^2_{2,\alpha} * \sqrt{|\Sigma|} = [18.8 \quad 8.71 \quad 4.36]$

for $\alpha = [0.05 \quad 0.25 \quad 0.5]$

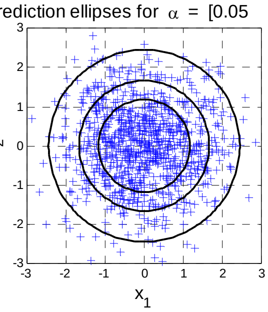
Probability density contour plot for $f(x_1,x_2)$
and ellipse eigenvector axes $\sqrt{\lambda_i} * e_i$ (shown for c^2=1)



3D plot of probability density function $f(x_1, x_2)$



N = 1000 observations scatter plot for (X_1, X_2) and
$(1-\alpha)$*100% prediction ellipses for $\alpha = [0.05 \quad 0.25 \quad 0.5]$



Bivariate normal distribution parameters and "geometry"

$(X_1,X_2) \sim N(\mu,\Sigma)$

$\mu = [0, 0]$ $\quad [\sigma_1, \sigma_2] = [1, 2]$ $\quad \rho = 0$

$\Sigma = [1\ \ 0; 0\ \ 4]$

total variation $= \Sigma_i \lambda_i = \sigma_1^2 + \sigma_2^2 = 5$

generalized variance $= |\Sigma| = \Pi_i \lambda_i = \sigma_1^2 * \sigma_2^2 * (1 - \rho^2) = 4$

---

constant density contour ellipses $(x-\mu)^T * \Sigma^{-1} * (x-\mu) = c^2$

eigenvalues and vectors for Σ $\quad \lambda_1 = 1 \quad e_1 = [1\ \ 0]$

$\lambda_2 = 4 \quad e_2 = [0\ \ 1]$

Mahalanobis (statistical) distance $(x-\mu)^T * \Sigma^{-1} * (x-\mu) \sim \chi^2_2$

---

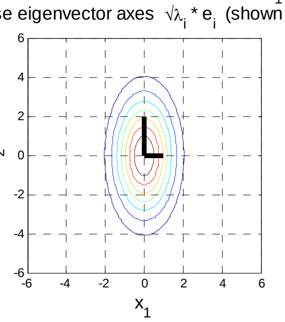
therefore $c^2 = \chi^2_{2,\alpha}$ gives $(1-\alpha)$*100% prediction ellipses for observations, x_i

ellipse half-length axes $\sqrt{\lambda_i} * \sqrt{\chi^2_{2,\alpha}} * e_i = [1 * e_1,\ 2 * e_2] * \sqrt{\chi^2_{2,\alpha}}$

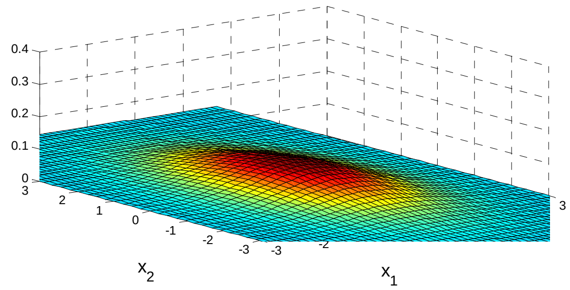
ellipse areas $A_\alpha = \pi * \chi^2_{2,\alpha} * \sqrt{|\Sigma|} = [37.6 \quad 17.4 \quad 8.71]$

for $\alpha = [0.05 \quad 0.25 \quad 0.5]$

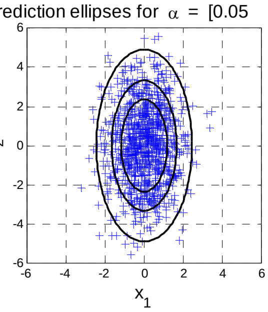
Probability density contour plot for $f(x_1,x_2)$
and ellipse eigenvector axes $\sqrt{\lambda_i} * e_i$ (shown for c^2=1)



3D plot of probability density function $f(x_1, x_2)$



N = 1000 observations scatter plot for (X_1, X_2) and
$(1-\alpha)$*100% prediction ellipses for $\alpha = [0.05 \quad 0.25 \quad 0.5]$



Bivariate normal distribution parameters and "geometry"

$(X_1,X_2) \sim N(\mu,\Sigma)$

$\mu = [0, 0]$ $[\sigma_1, \sigma_2] = [1, 1]$ $\rho = 0.7$

$\Sigma = [1\ \ 0.7; 0.7\ \ 1]$

total variation $= \Sigma_i \lambda_i = \sigma_1^2 + \sigma_2^2 = 2$

generalized variance $= |\Sigma| = \Pi_i \lambda_i = \sigma_1^2 * \sigma_2^2 * (1 - \rho^2) = 0.51$

---

constant density contour ellipses $(x\text{-}\mu)^T * \Sigma^{-1} * (x\text{-}\mu) = c^2$

eigenvalues and vectors for Σ $\lambda_1 = 0.3$ $e_1 = [-0.707\ \ 0.707]$

$\lambda_2 = 1.7$ $e_2 = [0.707\ \ 0.707]$

Mahalanobis (statistical) distance $(x\text{-}\mu)^T * \Sigma^{-1} * (x\text{-}\mu) \sim \chi_2^2$

---

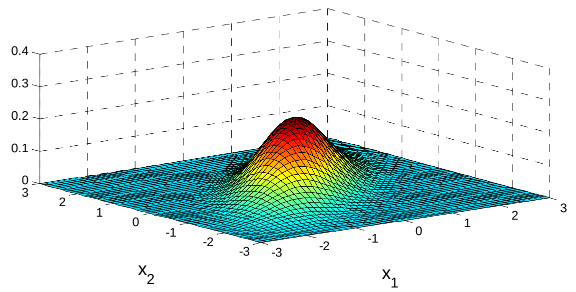
therefore $c^2 = \chi^2_{2,\alpha}$ gives $(1\text{-}\alpha)$*100% prediction ellipses for observations, x_i

ellipse half-length axes $\sqrt{\lambda_i} * \sqrt{\chi^2_{2,\alpha}} * e_i = [0.548 * e_1,\ 1.3 * e_2] * \sqrt{\chi^2_{2,\alpha}}$

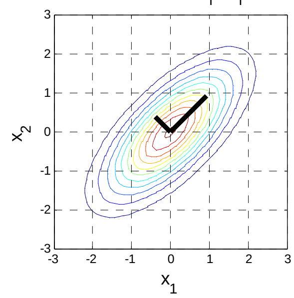
ellipse areas $A_\alpha = \pi * \chi^2_{2,\alpha} * \sqrt{|\Sigma|} = [13.4\quad 6.22\quad 3.11]$

for $\alpha = [0.05\quad 0.25\quad 0.5]$

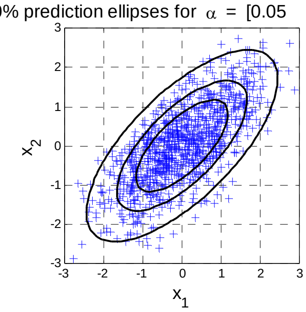
3D plot of probability density function $f(x_1, x_2)$



Probability density contour plot for $f(x_1,x_2)$
and ellipse eigenvector axes $\sqrt{\lambda_i} * e_i$ (shown for $c^2=1$)



N = 1000 observations scatter plot for (X_1, X_2) and
$(1\text{-}\alpha)$*100% prediction ellipses for $\alpha = [0.05\quad 0.25\quad 0.5]$



Bivariate normal distribution parameters and "geometry"

$(X_1,X_2) \sim N(\mu,\Sigma)$

$\mu = [0, 0]$ $[\sigma_1, \sigma_2] = [1, 1]$ $\rho = 0.9$

$\Sigma = [1\ \ 0.9; 0.9\ \ 1]$

total variation $= \Sigma_i \lambda_i = \sigma_1^2 + \sigma_2^2 = 2$

generalized variance $= |\Sigma| = \Pi_i \lambda_i = \sigma_1^2 * \sigma_2^2 * (1 - \rho^2) = 0.19$

---

constant density contour ellipses $(x\text{-}\mu)^T * \Sigma^{-1} * (x\text{-}\mu) = c^2$

eigenvalues and vectors for Σ $\lambda_1 = 0.1$ $e_1 = [-0.707\ \ 0.707]$

$\lambda_2 = 1.9$ $e_2 = [0.707\ \ 0.707]$

Mahalanobis (statistical) distance $(x\text{-}\mu)^T * \Sigma^{-1} * (x\text{-}\mu) \sim \chi_2^2$

---

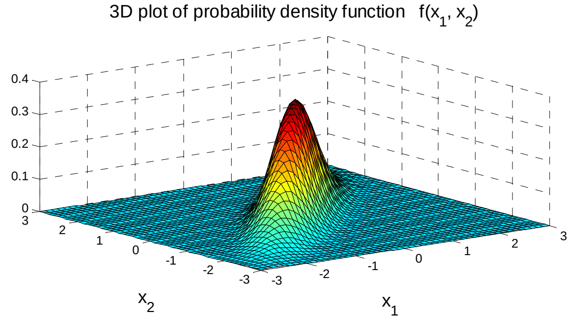
therefore $c^2 = \chi^2_{2,\alpha}$ gives $(1\text{-}\alpha)$*100% prediction ellipses for observations, x_i

ellipse half-length axes $\sqrt{\lambda_i} * \sqrt{\chi^2_{2,\alpha}} * e_i = [0.316 * e_1,\ 1.38 * e_2] * \sqrt{\chi^2_{2,\alpha}}$

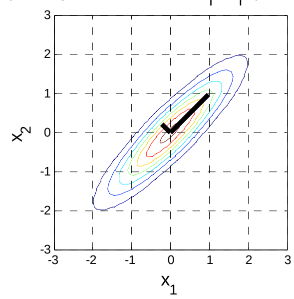
ellipse areas $A_\alpha = \pi * \chi^2_{2,\alpha} * \sqrt{|\Sigma|} = [8.2\quad 3.8\quad 1.9]$

for $\alpha = [0.05\quad 0.25\quad 0.5]$

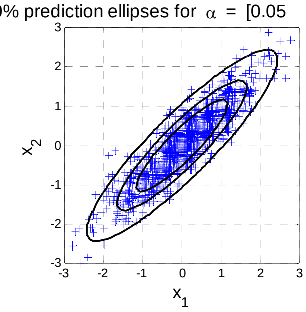
3D plot of probability density function $f(x_1, x_2)$



Probability density contour plot for $f(x_1,x_2)$
and ellipse eigenvector axes $\sqrt{\lambda_i} * e_i$ (shown for $c^2=1$)



N = 1000 observations scatter plot for (X_1, X_2) and
$(1\text{-}\alpha)$*100% prediction ellipses for $\alpha = [0.05\quad 0.25\quad 0.5]$



Bivariate normal distribution parameters and "geometry"

$(X_1,X_2) \sim N(\mu,\Sigma)$

$\mu = [0, 0]$ $[\sigma_1, \sigma_2] = [1, 2]$ $\rho = 0.7$

$\Sigma = [1\ \ 1.4;\ 1.4\ \ 4]$

total variation $= \Sigma_i \lambda_i = \sigma_1^2 + \sigma_2^2 = 5$

generalized variance $= |\Sigma| = \Pi_i \lambda_i = \sigma_1^2 * \sigma_2^2 * (1 - \rho^2) = 2.04$

---

constant density contour ellipses $(x-\mu)^T * \Sigma^{-1} * (x-\mu) = c^2$

eigenvalues and vectors for Σ $\lambda_1 = 0.44817$ $e_1 = [-0.93\ \ 0.367]$

$\lambda_2 = 4.5518$ $e_2 = [0.367\ \ 0.93]$

Mahalanobis (statistical) distance $(x-\mu)^T * \Sigma^{-1} * (x-\mu) \sim \chi_2^2$

---

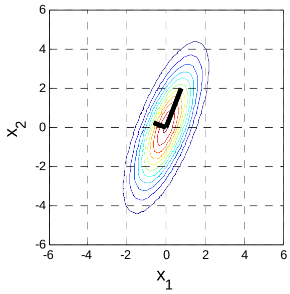
therefore $c^2 = \chi^2_{2,\alpha}$ gives $(1-\alpha)$*100% prediction ellipses for observations, x_i

ellipse half-length axes $\sqrt{\lambda_i} * \sqrt{\chi^2_{2,\alpha}} * e_i = [0.669 * e_1,\ 2.13 * e_2] * \sqrt{\chi^2_{2,\alpha}}$

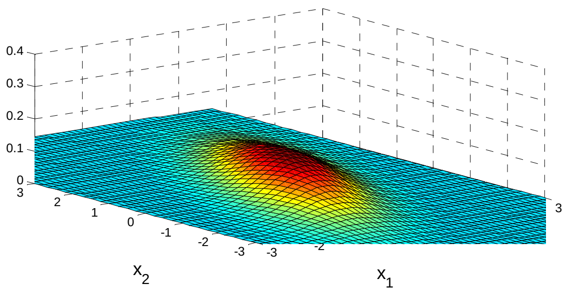
ellipse areas $A_\alpha = \pi * \chi^2_{2,\alpha} * \sqrt{|\Sigma|} = [26.9\ \ \ 12.4\ \ \ 6.22]$

for $\alpha = [0.05\ \ \ 0.25\ \ \ 0.5]$

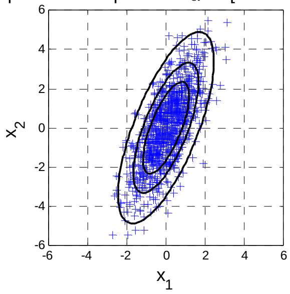
Probability density contour plot for $f(x_1,x_2)$
and ellipse eigenvector axes $\sqrt{\lambda_i} * e_i$ (shown for c^2=1)



3D plot of probability density function $f(x_1, x_2)$



N = 1000 observations scatter plot for (X_1, X_2) and
$(1-\alpha)$*100% prediction ellipses for $\alpha = [0.05\ \ \ 0.25\ \ \ 0.5]$



Bivariate normal distribution parameters and "geometry"

$(X_1,X_2) \sim N(\mu,\Sigma)$

$\mu = [0, 0]$ $[\sigma_1, \sigma_2] = [1, 2]$ $\rho = 0.9$

$\Sigma = [1\ \ 1.8;\ 1.8\ \ 4]$

total variation $= \Sigma_i \lambda_i = \sigma_1^2 + \sigma_2^2 = 5$

generalized variance $= |\Sigma| = \Pi_i \lambda_i = \sigma_1^2 * \sigma_2^2 * (1 - \rho^2) = 0.76$

---

constant density contour ellipses $(x-\mu)^T * \Sigma^{-1} * (x-\mu) = c^2$

eigenvalues and vectors for Σ $\lambda_1 = 0.15693$ $e_1 = [-0.906\ \ 0.424]$

$\lambda_2 = 4.8431$ $e_2 = [0.424\ \ 0.906]$

Mahalanobis (statistical) distance $(x-\mu)^T * \Sigma^{-1} * (x-\mu) \sim \chi_2^2$

---

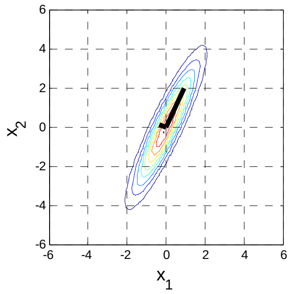
therefore $c^2 = \chi^2_{2,\alpha}$ gives $(1-\alpha)$*100% prediction ellipses for observations, x_i

ellipse half-length axes $\sqrt{\lambda_i} * \sqrt{\chi^2_{2,\alpha}} * e_i = [0.396 * e_1,\ 2.2 * e_2] * \sqrt{\chi^2_{2,\alpha}}$

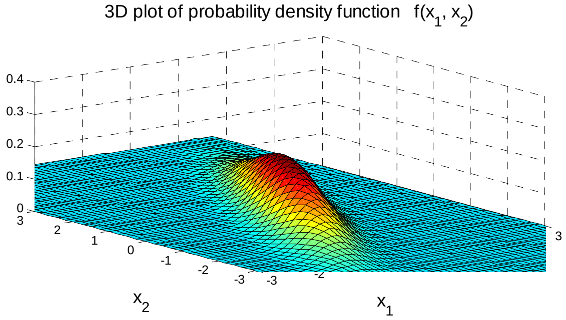
ellipse areas $A_\alpha = \pi * \chi^2_{2,\alpha} * \sqrt{|\Sigma|} = [16.4\ \ \ 7.59\ \ \ 3.8]$

for $\alpha = [0.05\ \ \ 0.25\ \ \ 0.5]$

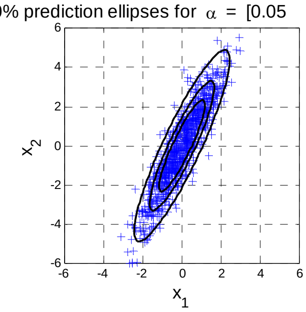
Probability density contour plot for $f(x_1,x_2)$
and ellipse eigenvector axes $\sqrt{\lambda_i} * e_i$ (shown for c^2=1)



3D plot of probability density function $f(x_1, x_2)$



N = 1000 observations scatter plot for (X_1, X_2) and
$(1-\alpha)$*100% prediction ellipses for $\alpha = [0.05\ \ \ 0.25\ \ \ 0.5]$



Bivariate normal distribution parameters and "geometry"

$(X_1,X_2) \sim N(\mu,\Sigma)$

$\mu = [0, 0]$ $[\sigma_1, \sigma_2] = [1, 1]$ $\rho = -0.7$

$\Sigma = [1\ \ -0.7; -0.7\ \ 1]$

total variation $= \Sigma_i \lambda_i = \sigma_1^2 + \sigma_2^2 = 2$

generalized variance $= |\Sigma| = \Pi_i \lambda_i = \sigma_1^2 * \sigma_2^2 * (1 - \rho^2) = 0.51$

---

constant density contour ellipses $(x-\mu)^T * \Sigma^{-1} * (x-\mu) = c^2$

eigenvalues and vectors for Σ $\lambda_1 = 0.3$ $e_1 = [-0.707\ \ -0.707]$

$\lambda_2 = 1.7$ $e_2 = [-0.707\ \ 0.707]$

Mahalanobis (statistical) distance $(x-\mu)^T * \Sigma^{-1} * (x-\mu) \sim \chi_2^2$

---

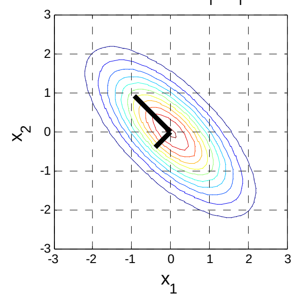
therefore $c^2 = \chi^2_{2,\alpha}$ gives $(1-\alpha)$*100% prediction ellipses for observations, x_i

ellipse half-length axes $\sqrt{\lambda_i} * \sqrt{\chi^2_{2,\alpha}} * e_i = [0.548 * e_1,\ 1.3 * e_2] * \sqrt{\chi^2_{2,\alpha}}$

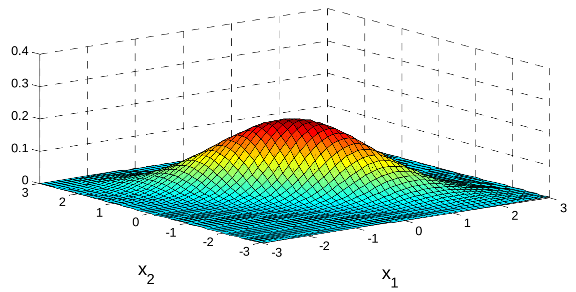
ellipse areas $A_\alpha = \pi * \chi^2_{2,\alpha} * \sqrt{|\Sigma|} = [13.4\ \ 6.22\ \ 3.11]$

for $\alpha = [0.05\ \ 0.25\ \ 0.5]$

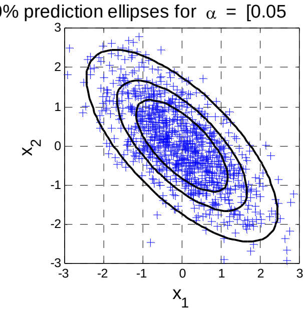
Probability density contour plot for $f(x_1,x_2)$
and ellipse eigenvector axes $\sqrt{\lambda_i} * e_i$ (shown for $c^2=1$)



3D plot of probability density function $f(x_1, x_2)$



N = 1000 observations scatter plot for (X_1, X_2) and
$(1-\alpha)$*100% prediction ellipses for $\alpha = [0.05\ \ 0.25\ \ 0.5]$



Bivariate normal distribution parameters and "geometry"

$(X_1,X_2) \sim N(\mu,\Sigma)$

$\mu = [0, 0]$ $[\sigma_1, \sigma_2] = [1, 1]$ $\rho = -0.9$

$\Sigma = [1\ \ -0.9; -0.9\ \ 1]$

total variation $= \Sigma_i \lambda_i = \sigma_1^2 + \sigma_2^2 = 2$

generalized variance $= |\Sigma| = \Pi_i \lambda_i = \sigma_1^2 * \sigma_2^2 * (1 - \rho^2) = 0.19$

---

constant density contour ellipses $(x-\mu)^T * \Sigma^{-1} * (x-\mu) = c^2$

eigenvalues and vectors for Σ $\lambda_1 = 0.1$ $e_1 = [-0.707\ \ -0.707]$

$\lambda_2 = 1.9$ $e_2 = [-0.707\ \ 0.707]$

Mahalanobis (statistical) distance $(x-\mu)^T * \Sigma^{-1} * (x-\mu) \sim \chi_2^2$

---

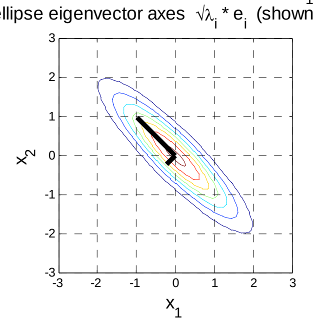
therefore $c^2 = \chi^2_{2,\alpha}$ gives $(1-\alpha)$*100% prediction ellipses for observations, x_i

ellipse half-length axes $\sqrt{\lambda_i} * \sqrt{\chi^2_{2,\alpha}} * e_i = [0.316 * e_1,\ 1.38 * e_2] * \sqrt{\chi^2_{2,\alpha}}$

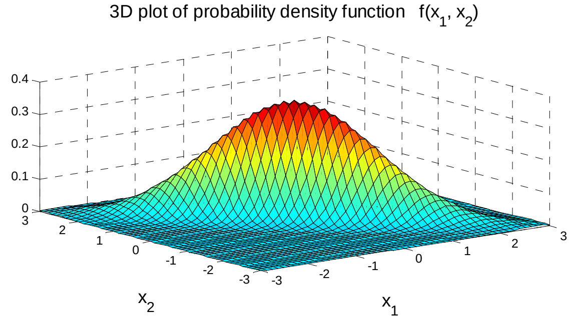
ellipse areas $A_\alpha = \pi * \chi^2_{2,\alpha} * \sqrt{|\Sigma|} = [8.2\ \ 3.8\ \ 1.9]$

for $\alpha = [0.05\ \ 0.25\ \ 0.5]$

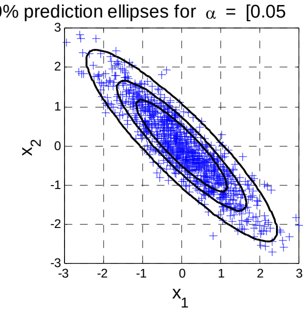
Probability density contour plot for $f(x_1,x_2)$
and ellipse eigenvector axes $\sqrt{\lambda_i} * e_i$ (shown for $c^2=1$)



3D plot of probability density function $f(x_1, x_2)$



N = 1000 observations scatter plot for (X_1, X_2) and
$(1-\alpha)$*100% prediction ellipses for $\alpha = [0.05\ \ 0.25\ \ 0.5]$



Bivariate normal distribution parameters and "geometry"

$(X_1,X_2) \sim N(\mu,\Sigma)$

$\mu = [0, 0]$ $[\sigma_1, \sigma_2] = [1, 2]$ $\rho = -0.7$

$\Sigma = [1\ \ -1.4; -1.4\ \ 4]$

total variation $= \Sigma_i \lambda_i = \sigma_1^2 + \sigma_2^2 = 5$

generalized variance $= |\Sigma| = \Pi_i \lambda_i = \sigma_1^2 * \sigma_2^2 * (1 - \rho^2) = 2.04$

---

constant density contour ellipses $(x-\mu)^T * \Sigma^{-1} * (x-\mu) = c^2$

eigenvalues and vectors for Σ $\lambda_1 = 0.44817$ $e_1 = [-0.93\ \ -0.367]$

$\lambda_2 = 4.5518$ $e_2 = [-0.367\ \ 0.93]$

Mahalanobis (statistical) distance $(x-\mu)^T * \Sigma^{-1} * (x-\mu) \sim \chi^2_2$

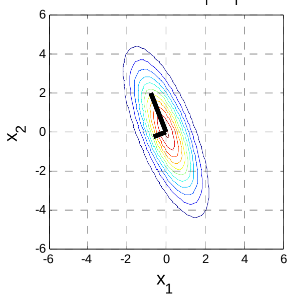
---

therefore $c^2 = \chi^2_{2,\alpha}$ gives $(1-\alpha)$*100% prediction ellipses for observations, x_i

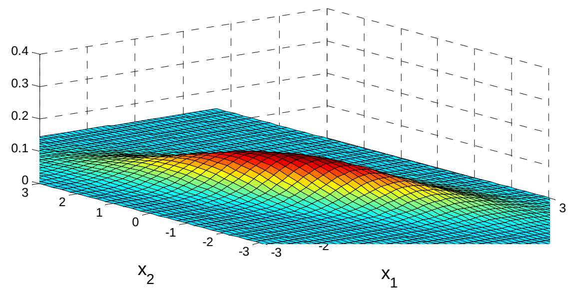
ellipse half-length axes $\sqrt{\lambda_i} * \sqrt{\chi^2_{2,\alpha}} * e_i = [0.669 * e_1,\ 2.13 * e_2] * \sqrt{\chi^2_{2,\alpha}}$

ellipse areas $A_\alpha = \pi * \chi^2_{2,\alpha} * \sqrt{|\Sigma|} = [26.9\ \ 12.4\ \ 6.22]$ for $\alpha = [0.05\ \ 0.25\ \ 0.5]$

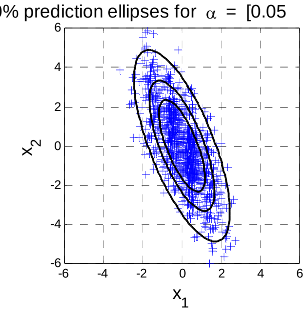
Probability density contour plot for $f(x_1,x_2)$ and ellipse eigenvector axes $\sqrt{\lambda_i} * e_i$ (shown for c^2=1)



3D plot of probability density function $f(x_1, x_2)$



N = 1000 observations scatter plot for (X_1, X_2) and $(1-\alpha)$*100% prediction ellipses for $\alpha = [0.05\ \ 0.25\ \ 0.5]$



Bivariate normal distribution parameters and "geometry"

$(X_1,X_2) \sim N(\mu,\Sigma)$

$\mu = [0, 0]$ $[\sigma_1, \sigma_2] = [1, 2]$ $\rho = -0.9$

$\Sigma = [1\ \ -1.8; -1.8\ \ 4]$

total variation $= \Sigma_i \lambda_i = \sigma_1^2 + \sigma_2^2 = 5$

generalized variance $= |\Sigma| = \Pi_i \lambda_i = \sigma_1^2 * \sigma_2^2 * (1 - \rho^2) = 0.76$

---

constant density contour ellipses $(x-\mu)^T * \Sigma^{-1} * (x-\mu) = c^2$

eigenvalues and vectors for Σ $\lambda_1 = 0.15693$ $e_1 = [-0.906\ \ -0.424]$

$\lambda_2 = 4.8431$ $e_2 = [-0.424\ \ 0.906]$

Mahalanobis (statistical) distance $(x-\mu)^T * \Sigma^{-1} * (x-\mu) \sim \chi^2_2$

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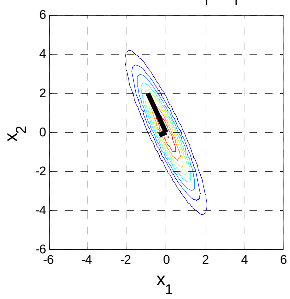
therefore $c^2 = \chi^2_{2,\alpha}$ gives $(1-\alpha)$*100% prediction ellipses for observations, x_i

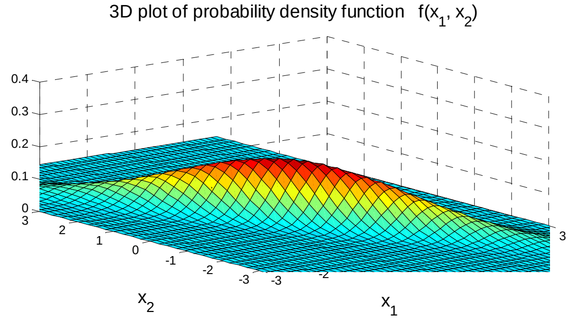
ellipse half-length axes $\sqrt{\lambda_i} * \sqrt{\chi^2_{2,\alpha}} * e_i = [0.396 * e_1,\ 2.2 * e_2] * \sqrt{\chi^2_{2,\alpha}}$

ellipse areas $A_\alpha = \pi * \chi^2_{2,\alpha} * \sqrt{|\Sigma|} = [16.4\ \ 7.59\ \ 3.8]$ for $\alpha = [0.05\ \ 0.25\ \ 0.5]$

Probability density contour plot for $f(x_1,x_2)$ and ellipse eigenvector axes $\sqrt{\lambda_i} * e_i$ (shown for c^2=1)



3D plot of probability density function $f(x_1, x_2)$



N = 1000 observations scatter plot for (X_1, X_2) and $(1-\alpha)$*100% prediction ellipses for $\alpha = [0.05\ \ 0.25\ \ 0.5]$